

kafka-isotope Library

kafka-isotope provides **end-to-end record tracing for Apache Kafka** using Kafka record headers, a **ProducerInterceptor**, and optional Prometheus metrics.

It works by attaching lightweight tracing artifacts—called *isotopes*—to records as they move through Kafka pipelines.

An isotope is a small tracing payload carried in Kafka record headers. Like a biochemical isotope used to trace molecules through a metabolic pathway, it enables the journey of a record through an event-driven architecture to be observed and analyzed.

On first produce, **kafka-isotope** stamps a UUIDv7 trace ID and origin metadata onto the record. Each subsequent re-produce appends another hop, allowing the full path of a record through Kafka pipelines to be reconstructed.

Built entirely on Kafka's public extension points—a **ProducerInterceptor** plus record headers—**kafka-isotope** requires:

- No broker changes
- No sidecar agents
- No vendor lock-in

From isotope headers, **kafka-isotope** derives trace data that can be analyzed directly or exported as Prometheus metrics to support:

- End-to-end latency
- Pipeline topology
- Hop distribution
- Drift detection
- Stuck trace detection
- Per-trace forensic replay

This makes **kafka-isotope** a lightweight but powerful observability layer for Kafka-based event-driven architectures.

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1.0 Modules

Module	Coordinate	Pulls	Use it for
kafka-isotope-core	ai.signalroom:kafka-isotope-core	Jackson, SLF4J (kafka-clients is compileOnly)	Trace propagation — interceptor, headers, consume markers.
kafka-isotope-metrics	ai.signalroom:kafka-isotope-metrics	kafka-isotope-core + Micrometer/Prometheus	Optional /metrics exporter for the stateless reports.

kafka-isotope-core never depends on a metrics library: emission routes through a no-op `IsotopeMetricsSink` until kafka-isotope-metrics registers the Prometheus one, so propagation runs with zero metrics overhead.

2.0 Install using Gradle

```
repositories {
    mavenCentral()
}
dependencies {
    implementation 'ai.signalroom:kafka-isotope-core:0.18.0'
    implementation 'ai.signalroom:kafka-isotope-metrics:0.18.0' // optional
    - only for Prometheus
}
```

See [kafka-isotope-core/README.md](#) for the full adopter quickstart (registering the interceptor, adopting/marking on consume, starting the exporter).

3.0 Demo

A runnable reference pipeline (Confluent Platform / Confluent Cloud on Minikube, the seven Flink reports, and a one-command Prometheus + Grafana showcase) lives in the companion repo: [j3-signalroom/confluent-kafka-isotope](#).